

Ruthwik Reddy Bommana

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Profile Summary

- Software engineer with 4 years of experience building production systems end to end, with a core focus on **TypeScript**, **JavaScript**, **React**, and **Next.js** for high-traffic customer-facing applications.
- Build responsive interfaces using **HTML**, **CSS**, and accessibility best practices, including semantic markup, keyboard navigation, and careful focus management for complex form flows.
- Develop backend services in **Node.js** using frameworks like **Express** or **NestJS**, with REST API designs that cover validation, pagination, consistent error handling, and clear API documentation.
- Model data in **PostgreSQL** with relational schemas, constraints, and indexes, manage database migrations and transactions, and investigate slow queries while keeping concurrent updates and production changes safe using tools like **Prisma** or **Drizzle**.
- Operate services on cloud platforms with **Git**-based workflows, CI/CD pipelines, **Docker**-backed development environments, structured logging and monitoring, and end-to-end quality checks using **Vitest**, **Jest**, and **Playwright**.
- Use AI coding tools such as **Cursor**, **GitHub Copilot**, and **Claude Code** in a review-first way, focusing on correctness, security, and maintainability, and collaborate with stakeholders using **Figma** prototypes and clear technical explanations tied to business impact.

Technical Skills

Frontend: TypeScript, JavaScript, React, Next.js, React Hooks, React Query, Zustand, Redux Toolkit, React Router, Axios

UI & Styling: HTML5, CSS3, Tailwind CSS, Styled Components, Sass, PostCSS, Framer Motion

Accessibility & UX: ARIA, Headless UI, Radix UI, Storybook, Figma

Node & Backend: Node.js, Express, NestJS, Fastify, FastAPI, tRPC, REST APIs, OpenAPI (Swagger), Zod, Yup

Auth & Security: JWT, OAuth2, OpenID Connect, Auth0, Clerk, NextAuth, Helmet, OWASP ZAP

Databases: PostgreSQL, SQL, Prisma, Drizzle ORM, Knex, RDS (Aurora), pgAdmin

Caching & Queues: Redis, BullMQ, RabbitMQ, SQS

AWS: EC2, ECS (Fargate), EKS, Lambda, API Gateway, S3, CloudFront, RDS (Aurora), CloudWatch, AWS WAF

Azure: App Service, Azure Functions, Azure API Management, Azure Cosmos DB, Azure SQL Database, Azure Storage (Blob), Azure Key Vault, Application Insights, Azure Monitor, Azure DevOps Pipelines

Containers & CI: Docker, Docker Compose, Git, GitHub Actions, GitLab CI, CircleCI

Monitoring & Quality: Sentry, Datadog, New Relic, Grafana, Prometheus, Logstash

Testing: Vitest, Jest, Playwright, React Testing Library, Cypress, Supertest

AI & Coding Assistants: Cursor, Claude Code, GitHub Copilot, OpenAI API, Anthropic API

Integrations & Analytics: Stripe, HubSpot, Salesforce, Segment, Mixpanel, PostHog

Work Experience

Amazon

Boston, MA

Software Engineer

Jan 2024 – Present

- Refactored **Next.js** route boundaries and shared **React** components, tightening server and client responsibility across checkout and catalog flows.
- Orchestrated **React** hooks, route transitions, and async loading paths around **Next.js** rendering layers, keeping state consistent through tab changes, refreshes, and partial hydration.
- Eliminated form drift by redesigning validation, caching, and error states for asynchronous product entry flows that had been dropping responses under rapid user edits.
- Reworked a legacy **Node.js** service layer into typed **TypeScript** REST endpoints, standardizing pagination, request validation, and error handling for downstream consumers.
- Redesigned webhook processing to be idempotent with retry routing and dead letter handling, preserving **PostgreSQL** transactional integrity during duplicate deliveries and concurrent updates.
- Rebuilt authentication and role checks around third-party account links, enforcing permission gates through server-side policies before sensitive actions reached the UI.

- Removed brittle rendering paths from shared storefront components, splitting client-only interactions away from server-rendered data so the page stopped re-requesting unchanged payloads.
- Simplified an inherited dependency graph in the checkout stack, extracting reusable helpers and tightening contracts between form state, API calls, and display logic.
- Optimized responsive layouts with semantic **HTML** and structured **CSS**, keeping typography, spacing, and focus order stable across desktop, tablet, and mobile breakpoints.
- Constrained bundle growth by trimming unused code paths, correcting import boundaries, and cutting a 2.4 MB frontend build that had been slowing first render.
- Instrumented key flows with structured logs, correlation ids, and error tracking, then tied the signals to **CloudWatch** alarms for faster review of production regressions.
- Deployed background workers on **ECS Fargate** and burst handlers on **Lambda**, smoothing retry-heavy jobs without forcing the service tier to absorb queue spikes.
- Protected public APIs behind **API Gateway** and **AWS WAF**, keeping abuse checks close to the edge while preserving stable contracts for legitimate clients.
- Built unit, integration, and end-to-end coverage with **Vitest**, **Jest**, and **Playwright**, including browser cases for keyboard navigation, focus management, and failed submissions.
- Used **Cursor**, **GitHub Copilot**, **OpenAI**, and retrieval over **Vector DBs** to prototype edge cases quickly, then reviewed each generated path for correctness, security, and compatibility.

Fiserv

Hyderabad, India

Software Developer

Dec 2019 – Apr 2022

- Built secure banking workflows in **JavaScript** and **TypeScript** when the release window would not allow external dependencies, keeping the server-side modules and customer screens aligned without adding platform risk.
- Orchestrated request handling across **Node.js**, **TypeScript**, and **Azure API Management**, wiring consistent error normalization, session checks, and role-based permissions into the API path before anything reached downstream services.
- Resolved failing account flows by tracing validation gaps, async state drift, and duplicate submissions, then tightening the form pipeline with clearer loading states and **Azure Application Insights** traces for faster diagnosis.
- Integrated customer-facing journeys with responsive **HTML** and **CSS** across **Azure App Service**, balancing layout constraints, keyboard access, and field-level feedback for forms that had to behave cleanly on small screens and desktop browsers.
- Migrated schema changes through transactional database scripts in **PostgreSQL** and **Azure SQL Database**, adding constraints, safe rollbacks, and index updates while protecting live financial records during production changes.
- Improved slow ledger lookups by reading query plans, adding targeted SQL indexes, and trimming unused joins, which cut one high-volume reconciliation query from 1.8s to 240ms in production.
- Refactored aging endpoint code into smaller **TypeScript** modules and removed repeated parsing logic, leaving the API surface easier to review, easier to test, and less tied to one-off branch behavior.
- Documented and simplified the environment setup across **Azure Key Vault**, **Azure Storage (Blob)**, and local **Docker** workflows, replacing hand-managed secrets and brittle startup steps with reproducible configuration.
- Normalized REST responses around shared failure shapes and validation rules, then extended **Jest** coverage for integration paths that exercised authorization failures, database transactions, and browser-submitted forms.
- Deployed feature branches through **Azure DevOps Pipelines**, keeping pull requests, checks, and approvals tied to each release while moving configuration into named environments and secret stores.
- Monitored application behavior with **Azure Monitor** and **Azure Application Insights**, then used logs, request traces, and database plans together to isolate intermittent API latency before it spread across adjacent screens.

Certificates

AWS Certified Solutions Architect – Associate

Education

Clemson University

SC

Master of Science in Computer Science

Lovely Professional University

India

Bachelor of Technology in Computer Engineering